Topic 7: Design (part 2)

SCSM2113 Human-Computer Interaction Fundamentals
Faculty of Computing
Universiti Teknologi Malaysia

1

01 DESIGN PRINCIPLES

02 VISUAL PERCEPTION

03 INDUSTRY PRACTICE

2

HCI
PRINCIPLES

3

Shneiderman's 8 Golden Rules of Interface Design

The principles	Questions to consider	Mark Complete
1. Strive for consistency	Is the style of this element maintained across your site/app? Is this content placed in the correct location according to the site hierarchy? Does this follow the conventions for your chosen platform? How can you make your designs more consistent?	☐
2. Enable frequent users to use shortcuts	Are there shortcuts available for your more experienced users? Who is this product designed for? Will there be a need to consider experienced users? How can you make it easier and quicker for experienced users?	☐
3. Offer informative feedback	Does the user know where they are at in the process? Does the user know what they have done after performing this action? How are you communicating this feedback to your user?	☐
4. Design dialogue to yield closure	Does the user have to do any guessing here? Is it clear and obvious enough for your intended audience? Are there any next steps for the user? How are you communicating the system status with the user?	☐
5. Offer simple error handling	Have you done everything imaginable to prevent this error from happening on your end? Is this error avoidable in the first place? If the user does make an error, how easy is it for them to fix it?	☐
6. Permit easy reversal of actions	How many steps does the user have to take to reverse their actions? Will the user quickly realize they need to reverse the action in the first place? How can you make your users detect the possibility of reversal?	☐
7. Support internal locus of control	Will the user feel in control at this specific touch point in your app? Will they be surprised in an unpleasant manner? Does the site feel easily navigable? Does the user feel safe and in control? How can you make the user feel more safe and in control?	☐
8. Reduce short-term memory load	Are there enough visual cues here for the user to find the functionality or item? Do they have to remember things to understand what's going on? How can you help the user recall?	☐

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Norman's 7 Principles (1988):

1. Use both knowledge in the world and knowledge in the head.
2. Simplify the structure of tasks.
3. Make things visible.
4. Get the mappings right.
5. Exploit the power of constraints, both natural and artificial.
6. Design for error.
7. When all else fails, standardize.

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18 Principles of Interaction Design by Bruce Tognazzini

No	Principle	Description
1	Anticipation	The system should attempt to anticipate the user's needs.
2	Autonomy	The user should feel in control of the system and its actions.
3	Color Blindness	Avoid using color as the only means of conveying information.
4	Consistency	The interface should be consistent in appearance and behavior.
5	Defaults	Reasonable default values should be provided.
6	Efficiency of the User	Design for user efficiency, not just system performance.
7	Explorable Interfaces	Users should be able to explore safely without fear of error.
8	Fitts's Law	Larger and closer targets are easier to hit.
9	Human Interface Objects	Use familiar interface objects that reflect real-world analogies.
10	Latency Reduction	Minimize delays and provide feedback to make latency feel shorter.
11	Learnability	The interface should be easy to learn and remember.
12	Metaphors	Use metaphors that are familiar and intuitive to users.
13	Protect Users' Work	Users' work should be preserved, even in the event of errors.
14	Readability	Text should be legible and easy to scan.
15	Track State	Keep users informed of the system's current state.
16	Visible Navigation	Navigation options should be easily visible and accessible.
17	User Control	Users should initiate and control actions, not the system.
18	Visible Interfaces	Important elements should be visible without unnecessary clutter.

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VISUAL PERCEPTION

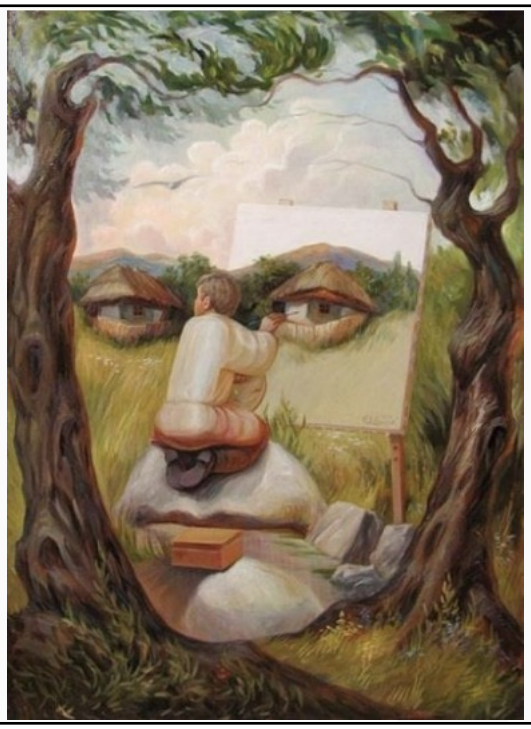
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7



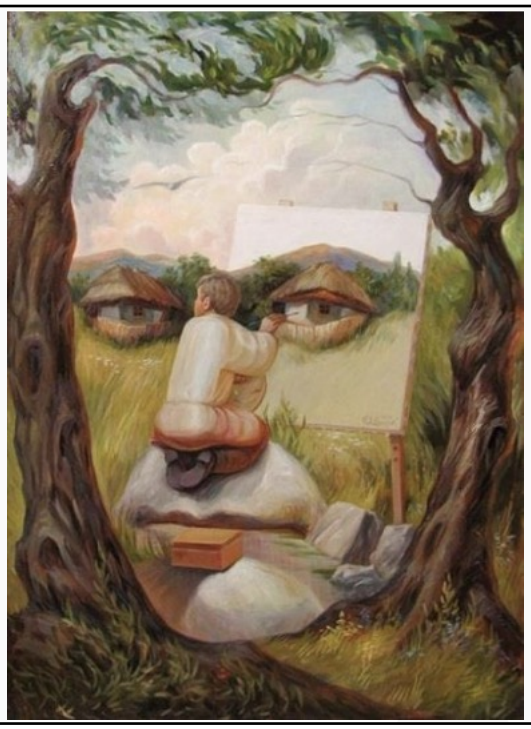
What is the ***first*** thing you see?

8



At first glance
you would see
a **human face**

9



But looking
more closely
you start to see
the details

10

The idea of Gestalt principles:

"The whole of something is more ***important*** to our understanding than the individual parts."

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Gestalt Principles in UI Design

- Similarity
- Proximity
- Enclosure
- Continuation
- Figure & Ground

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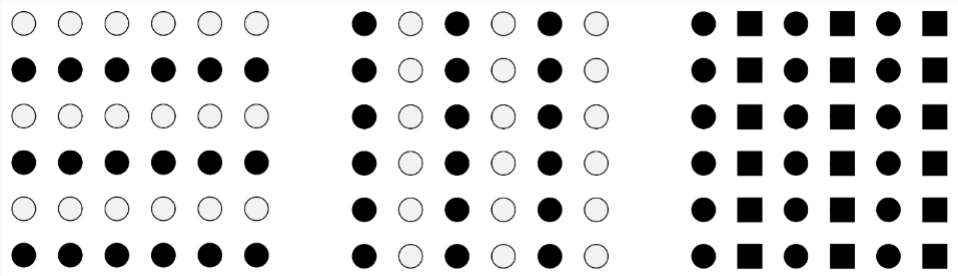
Similarity

Similar objects are seen as grouped

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Similarity

- Similarity of objects form groupings:
  - At left, light and dark shades create similarities among horizontal groups
  - At center, light and dark shades creates similarities among vertical groups
  - At right, circle and square shapes create similarities among vertical groups



Source:
http://newmediaabington.pbworks.com/w/page/66930745/Ges

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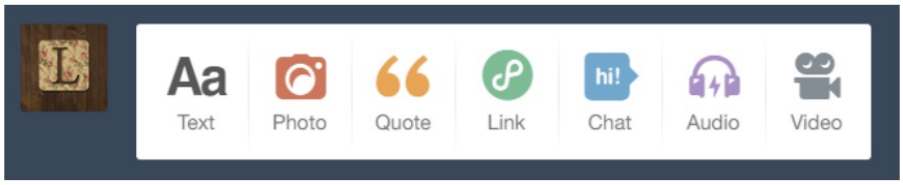
Similarity

- Each icon has a different image
- These icons are similar in shape and style
- They are perceived to be one group and share the same level of function



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Similarity



- Each block has a different icon
- The blocks share similar structure: an icon with text underneath
- They are perceived to be one group and share similar function: create a new blog post
- Users know exactly what UI elements to look for

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Proximity

When elements are *close together*, they are seen as grouped

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Proximity

By using white space, we can separate as well as build the relationship between elements

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Proximity



- Language & Currency fields are put close enough to be seen as one group
- The links on the right side are arranged with proper amount of white space. They are seen as 3 small groups and together form one big group

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Proximity

- White space, however, is just one part of implementing proximity.
- A design can have a lot of white space, but if items are not **grouped correctly**, the white space will have little effect.



Two business cards

Source: https://www.webdesignerdepot.com/2010/01/the-principle-of-proximity-in-web-design/

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Proximity

- Using white space effectively and grouping related elements are critical to giving your website a clear visual hierarchy.
- Hierarchy is conveyed by the way in which elements are grouped and sub-grouped.
- This hierarchy helps the user understand where they've been and where they want to go and, thus, helps to communicate the purpose of the website.
- A list is a good example of an element that has the potential to communicate visual hierarchy.



The list on the left has a clear visual hierarchy, showing the relationships between items (indented items are sub-categories of the primary items)

Source: https://www.webdesignerdepot.com/2010/01/the-principle-of-proximity-in-web-design/

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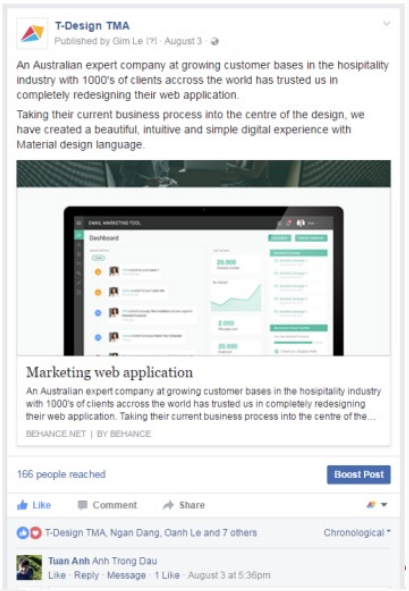
Enclosure

Things that have a *boundary* around them are seen as grouped

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Enclosure

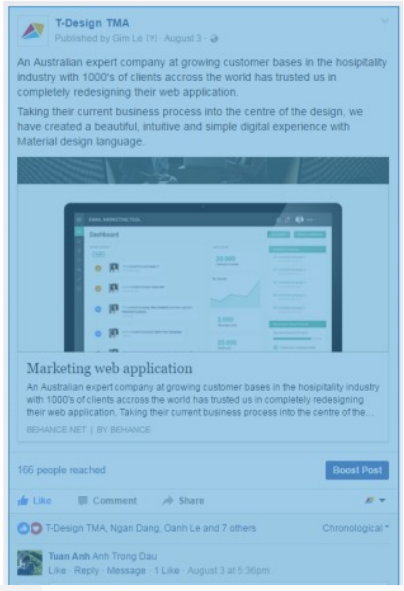
- Have you ever noticed the way Facebook elements are grouped together?



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Enclosure

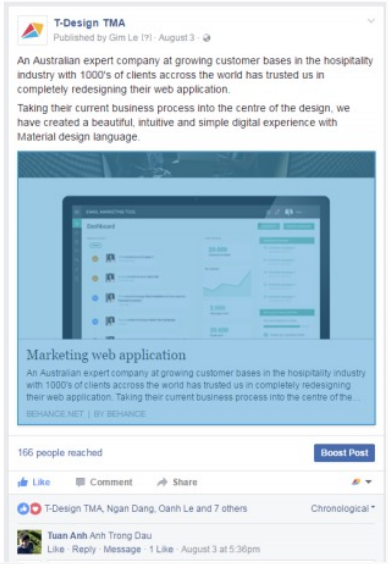
- There is a border around the whole post
- This is to separate one post to another



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Enclosure

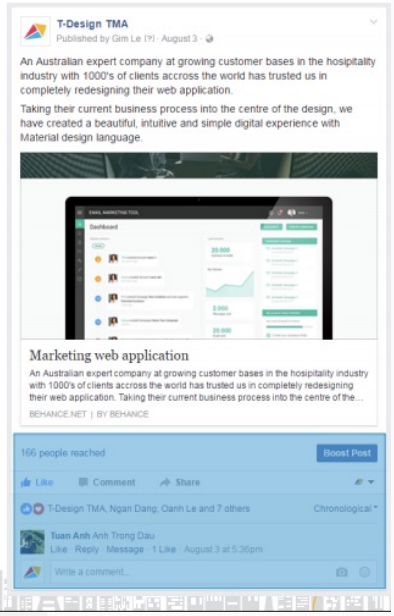
- There is another border around the link representation
- The image, the title along with a short description appear to be grouped



25

Enclosure

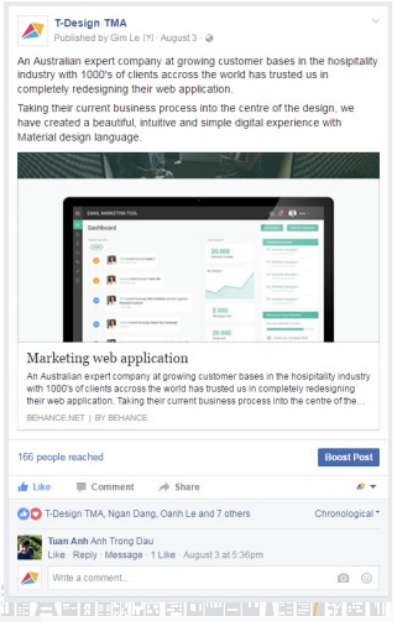
- This area at the bottom is for social interactions and is formed by several lines
- This area is also divided into different sections



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Enclosure

- Enclosure is very helpful when you have limited space with a lot of information



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Enclosure

To separate the link of the picture from the post

To separate one post with another posts

To separate the comment section for the related post



Source: https://www.facebook.com/univteknologimalaysia/

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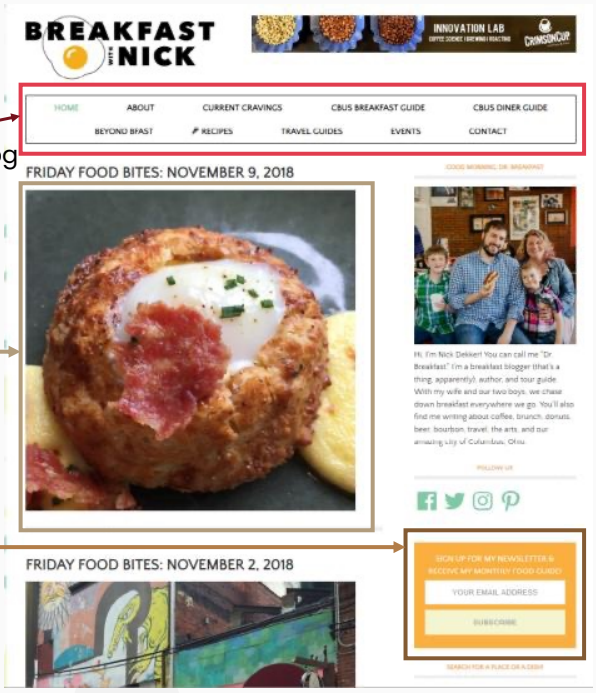
Enclosure

To separate the menu options available in the blog

To separate one post with another posts

To separate the link where users can subscribe monthly food guide by

Source: https://www.breakfastwithnick.com/



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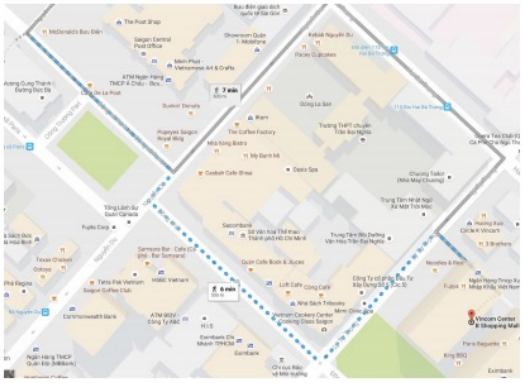
Continuation

Human eye follows *lines, curves*, or a *sequence of shapes*

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Continuation

Human eyes tend to see a single dotted line instead of a series of blue dots



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Continuation



- The light gray line represents the length of the track
- There is another dark gray line placed over the light gray line
- When the dark line grows, we understand the passage of time

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Continuation

- The viewer's eye will naturally follow a line or curve.
- The smooth flowing crossbar of the "H" leads the eye directly to the maple leaf.
- Principle of good continuity is when human perceive each of two or more objects as different, singular and uninterrupted object even when they intersect.
- The alignment of the objects or forms plays a major role for this principle to take effect.
- This principle is well used by educator to teach kids on how to write letters of the alphabet and draw images.





Sources: http://graphicdesign.spokanefalls.edu/tutorials/process/gestaltprinciples/gestaltprinc.htm
https://en.wikipedia.org/wiki/Principles_of_grouping

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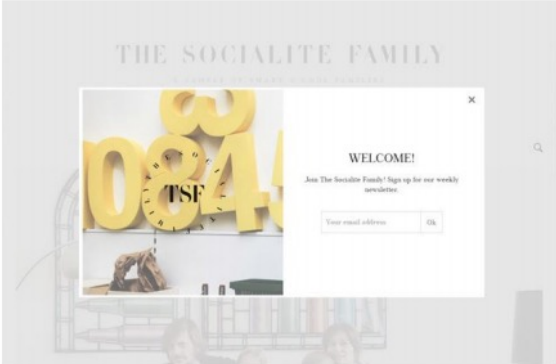
Figure & Ground

Elements are seen as either "*figures*" or "*grounds*".

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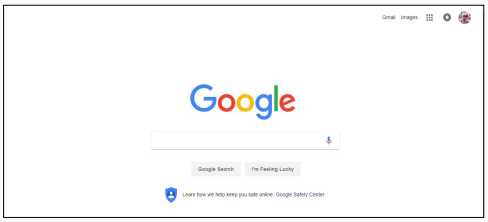
Figure & Ground

- Very common in modals
- Figure: the Welcome box
- Ground: the transparent background that softens the content in the back



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Figure & Ground



Example of **figure** (*the Google text and search bar)* and **ground** *(the white background)*



Example of **figure** (*bird*) and **ground** (*the surrounding trees*).

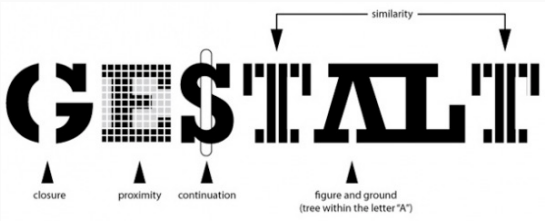
Sources:
https://wairimusensationndperception.weebly.com/figure-ground.html
https://www.google.com

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Example of Gestalt Principles Applied to Web Design

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Example of Gestalt Principles Applied to Web Design

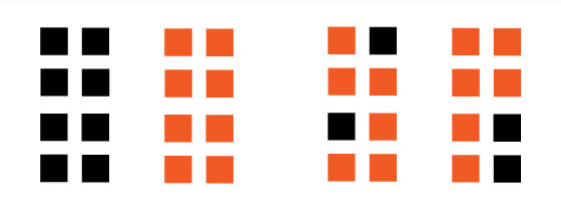


Reference: Gayle Christopher, gaylechristopher.com

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Proximity/Grouping

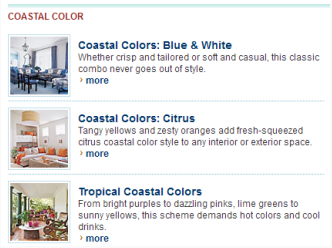
- Things that are close together are perceived to be more related than things that are spaced father apart.
- One of the first principles to impact perception
- Proximity overpowers similarity
- Only uniform connectedness overpowers proximity

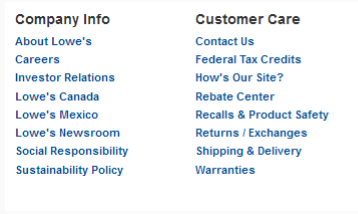


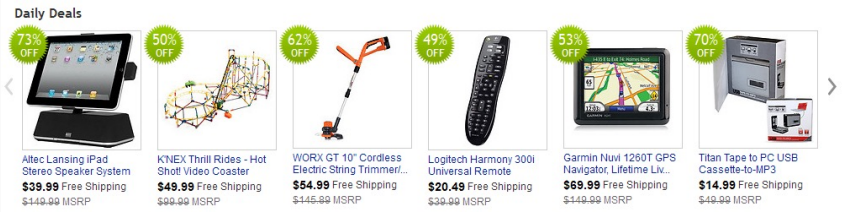
Proximity overpowers similarity in color/contrast

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Proximity/Grouping



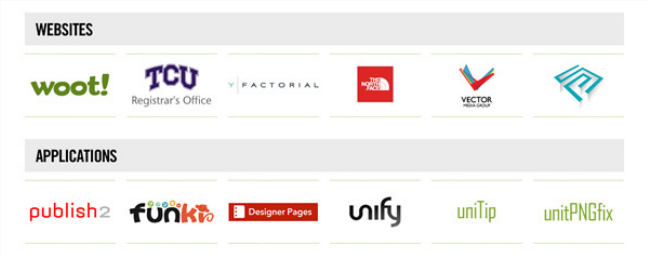




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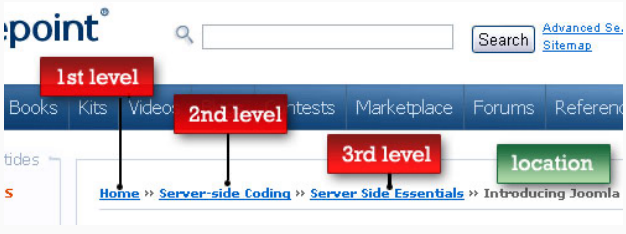
Good Continuation

- Elements arranged on a line or curve are perceived to be more related than elements not on the line or curve.
- All written language employs the principle of good continuation.
- Breadcrumbs, words in a paragraph, graphs, and linear arrangement (vertical or horizontal) communicate relatedness.



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Good Continuation

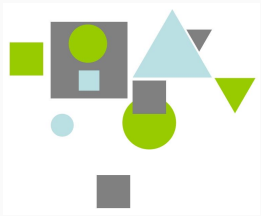




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Similarity

- Visual elements that are similar in shape, size, color and direction are perceived as part of a group.
- Different modes of similarity are not created equal.
- Color is the strongest way to suggest relationships.
- Used in links, icons, page content to suggest similar behavior, relatedness, and reinforce content hierarchies to communicate context.
- Consistency in styling and content dimension is important and implies structure.



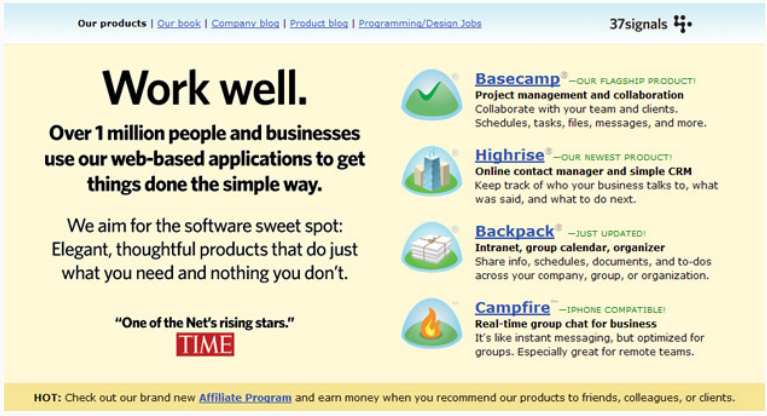


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Similarity

Mac OS

Blue Water



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Figure-Ground

- Elements are perceived as either figures (elements of focus) or background.
- Human mind must rapidly decide which elements to focus on in a scene. This ability allows us to determine what we should pay attention to and what we can ignore even if it does provide context.
- Use color, shading, highlights to cue the viewer in on what has focus and ground depth: buttons that look like press-able buttons, links that change on rollover, overlays that use shadow.

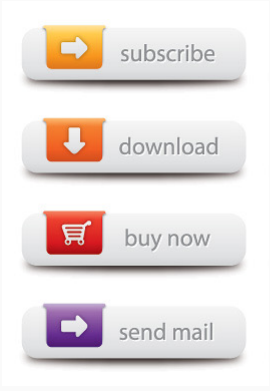
FedEx

Hope for African Children Initiative



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Figure-Ground





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Law of Prägnanz

- The law of Pragnanz—or principle of simplicity—is a fundamental Gestalt psychology law concerning how humans process visual information
- People usually perceive complex things as simplified forms to easily recognise and understand what they see
- Designers cater to this tendency to see things as the sum of simple parts in digital products
- It also goes by the names "the law of good Gestalt" and "the law of simplicity."

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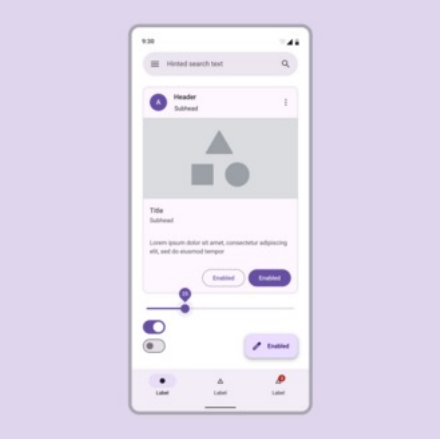
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INDUSTRY PRACTICES

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Google Material Design Principles



Principle	Description
Material is Metaphor	Uses tactile reality (shadows, surfaces, and edges) inspired by paper and ink to provide cues.
Bold, Graphic, Intentional	Emphasizes deliberate color choices, edge-to-edge imagery, large-scale typography, and intentional white space.
Motion Provides Meaning	Animations are meaningful and reinforce the user's actions, helping with orientation and continuity.
Adaptive Across Devices	Designs are responsive, scalable, and work seamlessly on all screen sizes and devices.
User Action Driven	Interfaces are focused on user input and promote direct manipulation with clear feedback.

Apple Human Interface Guideline

Principle	Description
Clarity	Ensure interfaces are easy to understand with legible text, precise icons, and straightforward navigation to eliminate ambiguity.
Deference	Minimize distractions by prioritizing content over UI elements, allowing users to focus on tasks with a clean, unobtrusive design.
Depth	Use visual hierarchy, layering, and subtle effects (e.g., shadows) to create a multi-dimensional experience that guides users naturally.
Consistency	Maintain familiar design patterns and behaviors across apps to align with Apple's ecosystem, enhancing user familiarity and comfort.
Feedback	Provide clear, immediate responses to user actions (e.g., visual cues, animations) to confirm interactions and reduce confusion.
Simplicity	Keep interfaces uncluttered, using minimal design elements and standard UI components (e.g., UIKit) to ensure intuitive functionality.